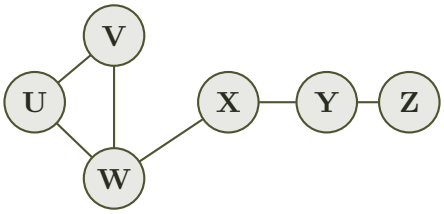
# Module 8: Graph Theory and Networks — Check-In

*Same core skills as the prequiz, with a different community network. Show enough work to make your thinking visible.*

Name: _____ Date: _____




The nodes U, V, W, X, Y, Z represent community centers and the edges are UV, UW, VW, WX, XY, YZ.

1. List W's direct neighbors and degree. Find the sum of all six node degrees, and check it against the number of edges.

2. Give one shortest path from U to Z and its length in edges. Explain why U can reach Z without a direct U–Z connection.

3. Remove W–X. List the resulting connected components. Can U still reach Z? Give a reason.

4. A grant allows one additional link: U–X or X–Z. Which proposal keeps U connected to Z if W–X later fails? Give a surviving path and explain why the other proposal does not work.

5. Start with only X informed in round 0. Informed locations send to neighbors each round; new recipients relay the following round. List the **newly informed nodes** in rounds 1 and 2. How many locations are informed in total after round 2?